1. Provide a list of movies where the words 'drama' and 'teacher' appear in the description using FTS

```
SELECT title, description
FROM film
WHERE MATCH(description) AGAINST('+drama +teacher' IN BOOLEAN MODE);
```

#	title	description
1	ACADEMY DINOSAUR	A Epic Drama of a Feminist And a Mad Scientist who must Battle a Teacher in The Canadian Rockies
2	ENCINO ELF	A Astounding Drama of a Feminist And a Teacher who must Confront a Husband in A Baloon
3	NUTS TIES	A Thoughtful Drama of a Explorer And a Womanizer who must Meet a Teacher in California
4	POND SEATTLE	A Stunning Drama of a Teacher And a Boat who must Battle a Feminist in Ancient China
5	RAIDERS ANTITRUST	A Amazing Drama of a Teacher And a Feminist who must Meet a Woman in The First Manned Space Station
6	RESERVOIR ADAPTATION	A Intrepid Drama of a Teacher And a Moose who must Kill a Car in California
7	SMILE EARRING	A Intrepid Drama of a Teacher And a Butler who must Build a Pastry Chef in Berlin
8	WAR NOTTING	A Boring Drama of a Teacher And a Sumo Wrestler who must Challenge a Secret Agent in The Canadian Rockies

#	Time	Action	Message	Duration / Fetch
47	09:10:14	select * from film_text order by film_id LIMIT 0, 1000	1000 row(s) returned	0.0010 sec / 0.0011 ...
48	09:10:20	select * from film_text order by film_id desc LIMIT 0, 1000	1000 row(s) returned	0.00073 sec / 0.002...
49	09:16:49	SELECT title, description FROM film WHERE MATCH(des...	Error Code: 1191. Can't find FULLTEXT index matc...	0.00042 sec
50	09:17:30	ALTER TABLE film ADD FULLTEXT(description)	0 row(s) affected, 1 warning(s): 124 InnoDB rebuilding table to add column FTS_D... Records: 0 Duplicates: 0 Warnings: 1	0.222 sec
51	09:18:04	SELECT title, description FROM film WHERE MATCH(des...	9 row(s) returned	0.0014 sec / 0.0000...

2. Add to the inventory table the is_available column of type Boolean or tinyint(1):
Creating a trigger on the rental table does the following:

- When the disc is rented, the is_available value is updated to false
- When the disc is returned, the is_available value is updated to true.

```
DELIMITER //

CREATE TRIGGER after_rental_insert
AFTER INSERT ON rental
FOR EACH ROW
BEGIN
    UPDATE inventory
    SET is_available = 0
    WHERE inventory_id = NEW.inventory_id;
END //

DELIMITER ;
```

#	Time	Action	Message	Duration / Fetch
50	09:17:30	ALTER TABLE film ADD FULLTEXT(description)	124 InnoDB rebuilding table to add column FTS_D... Records: 0 Duplicates: 0 Warnings: 1	0.222 sec
51	09:18:04	SELECT title, description FROM film WHERE MATCH(des...	9 row(s) returned	0.0014 sec / 0.0000...
52	09:19:16	ALTER TABLE inventory ADD COLUMN is_available TINYIN...	0 row(s) affected, 1 warning(s): 1681 Integer display width is deprecated and will ... Records: 0 Duplicates: 0 Warnings: 1	0.030 sec
53	09:19:48	CREATE TRIGGER after_rental_insert AFTER INSERT ON r...	0 row(s) affected	0.0082 sec

```
DELIMITER //

CREATE TRIGGER after_rental_update
AFTER UPDATE ON rental
FOR EACH ROW
BEGIN
    IF OLD.return_date IS NULL AND NEW.return_date IS NOT NULL THEN
        UPDATE inventory
        SET is_available = 1
        WHERE inventory_id = NEW.inventory_id;
    END IF;
END //

DELIMITER ;
```

#	Time	Action	Message	Duration / Fetch
			Records: 0 Duplicates: 0 Warnings: 1	
51	09:18:04	SELECT title, description FROM film WHERE MATCH(des...	9 row(s) returned	0.0014 sec / 0.0000...
52	09:19:16	ALTER TABLE inventory ADD COLUMN is_available TINYIN...	0 row(s) affected, 1 warning(s): 1681 Integer display width is deprecated and will ... Records: 0 Duplicates: 0 Warnings: 1	0.030 sec
53	09:19:48	CREATE TRIGGER after_rental_insert AFTER INSERT ON r...	0 row(s) affected	0.0082 sec
54	09:20:34	CREATE TRIGGER after_rental_update AFTER UPDATE O...	0 row(s) affected	0.0076 sec

```
UPDATE inventory
SET is_available = 0
WHERE inventory_id IN (
    SELECT inventory_id FROM rental WHERE return_date IS NULL
);
```

#	Time	Action	Message	Duration / Fetch
52	09:19:16	ALTER TABLE inventory ADD COLUMN is_available TINYIN...	0 row(s) affected, 1 warning(s): 1681 Integer display width is deprecated and will ... Records: 0 Duplicates: 0 Warnings: 1	0.030 sec
53	09:19:48	CREATE TRIGGER after_rental_insert AFTER INSERT ON r...	0 row(s) affected	0.0082 sec
54	09:20:34	CREATE TRIGGER after_rental_update AFTER UPDATE O...	0 row(s) affected	0.0076 sec
55	09:20:56	UPDATE inventory SET is_available = 0 WHERE inventory...	183 row(s) affected Rows matched: 183 Changed: 183 Warnings: 0	0.058 sec

TEST trigger

-kiem tra dia nay co san sang k

```
SELECT inventory_id, film_id, is_available
FROM inventory
WHERE inventory_id = 10;
```

#	inventory_ii	is_available
1	10	1
*	NULL	NULL

#	Time	Action	Message	Duration / Fetch
54	09:20:34	CREATE TRIGGER after_rental_update AFTER UPDATE O...	0 row(s) affected	0.0076 sec
55	09:20:56	UPDATE inventory SET is_available = 0 WHERE inventory...	183 row(s) affected Rows matched: 183 Changed: 183 Warnings: 0	0.058 sec
56	09:31:36	SELECT inventory_id, film_id, is_available FROM inventor...	1 row(s) returned	0.00060 sec / 0.000...
57	09:31:44	SELECT inventory_id, film_id, is_available FROM inventor...	1 row(s) returned	0.00051 sec / 0.000...
58	09:31:58	SELECT inventory_id, film_id, is_available FROM inventor...	1 row(s) returned	0.00058 sec / 0.000...
59	09:32:13	SELECT inventory_id, is_available FROM inventory WHE...	1 row(s) returned	0.00053 sec / 0.000...

- thue dia

```sql
INSERT INTO rental (rental_date, inventory_id, customer_id, staff_id)
VALUES (NOW(), 10, 1, 1);

-- Kiểm tra lại bảng inventory
SELECT inventory_id, is_available
FROM inventory
WHERE inventory_id = 10;
```

#	inventory_ii	is_available
1	10	0

#	Time	Action	Message	Duration / Fetch
56	09:31:36	SELECT inventory_id, film_id, is_available FROM inventor...	1 row(s) returned	0.00060 sec / 0.000...
57	09:31:44	SELECT inventory_id, film_id, is_available FROM inventor...	1 row(s) returned	0.00051 sec / 0.000...
58	09:31:58	SELECT inventory_id, film_id, is_available FROM inventor...	1 row(s) returned	0.00058 sec / 0.000...
59	09:32:13	SELECT inventory_id, is_available FROM inventory WHE...	1 row(s) returned	0.00053 sec / 0.000...
60	09:33:26	SELECT inventory_id, is_available FROM inventory WHE...	1 row(s) returned	0.00060 sec / 0.000...
61	09:33:47	SELECT inventory_id, is_available FROM inventory WHE...	1 row(s) returned	0.00049 sec / 0.000...

- tra dia

```sql
-- Cập nhật ngày trả cho lần thuê gần nhất của inventory_id = 10
UPDATE rental
SET return_date = NOW()
WHERE inventory_id = 10 AND return_date IS NULL;

-- Kiểm tra lại bảng inventory
SELECT inventory_id, is_available
FROM inventory
WHERE inventory_id = 10;
```

#	inventory_ii	is_available
1	10	1

#	Time	Action	Message	Duration / Fetch
57	09:31:44	SELECT inventory_id, film_id, is_available FROM inventor...	1 row(s) returned	0.00051 sec / 0.000...
58	09:31:58	SELECT inventory_id, film_id, is_available FROM inventor...	1 row(s) returned	0.00058 sec / 0.000...
59	09:32:13	SELECT inventory_id, is_available FROM inventory WHE...	1 row(s) returned	0.00053 sec / 0.000...
60	09:33:26	SELECT inventory_id, is_available FROM inventory WHE...	1 row(s) returned	0.00060 sec / 0.000...
61	09:33:47	SELECT inventory_id, is_available FROM inventory WHE...	1 row(s) returned	0.00049 sec / 0.000...
62	09:34:28	SELECT inventory_id, is_available FROM inventory WHE...	1 row(s) returned	0.00058 sec / 0.000...